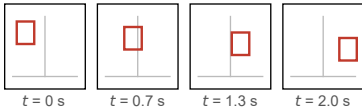


(a) Item

four frames over 2 s, target in a red box



“How fast is the player in the red box moving, in metres per second?”

Reference: tracked trajectory $\rightarrow$ court homography $\rightarrow$ speed (m/s) or path (m).

One game, 7 clips, 4,705 questions.

(b) Interventions (physical answer fixed)

Unit arms

same item asked in m, cm or rendered px;
exact factor 100 (cm), ≈ 27 (px)

Ruler arm

height cue replaced by “1 m spans ≈ 27 px”

First-frame $\times 4$

four copies of frame 1: same tokens,
no observed displacement

Text-only probe

convert stated numbers without images;
392 items, factor given or withheld

(c) Readings per cell

Rank agreement

Spearman ρ between predictions and
reference, per clip and family

Unit response

$R = \text{median}_i \hat{y}_i^{\text{unit}} / \hat{y}_i^{\text{metre}}$
(100 if the unit is applied)

Accuracy against a constant

T-MRA against the best single answer
chosen with the test labels

Degeneracy

≤ 2 distinct answers: the cell counts
towards no verdict